

No. of Functional Features Included in the Solution

Date	8 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status(Done / In Progress / Pending)	Marks Contribution
1	Interactive Input Form	Provides credit officers with clean input interfaces for key application features like income, loan amount, and credit history details.	User Interface (UI)	Done	1.0 Marks
2	Statistical Analysis Pipeline	Automatically executes Univariate, Bivariate, and Multivariate analysis structures to visualize demographic	Backend / Logic	Done	1.0 Marks

S.No	Feature Name	Feature Description	Module / Component	Status(Done / In Progress / Pending)	Marks Contribution
		distribution correlations.			
3	Automated Data Preprocessing	Handles runtime missing values through statistical imputation, handles class imbalances, and rescales features using uniform arrays.	Backend / Logic	Done	1.0 Marks
4	XGBoost Inference Engine	Routes valid application profiles to the serialized, top-performing machine learning model binary (\$81.1\%\$ test accuracy).	API / Integration	Done	1.0 Marks
5	Dynamic Status Alerts	Displays real-time decision flags on screen with instant visual validation alerts based on risk metrics.	User Interface (UI)	Done	1.0 Marks

Feature Summary:

Metric	Count / Value
Total Features Planned	5
Total Features Implemented	5
Core / Must-Have Features	4

Additional / Nice-to-Have Features	1
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Interactive Input Form, Dynamic Status Alerts
2	Backend / Logic	2	Statistical Analysis Pipeline, Automated Data Preprocessing
3	Database / Storage	0	(Local CSV Dataset used for offline model training)
4	API / Integration	1	XGBoost Inference Engine (Flask App Model Route)
5	Security / Authentication	2	(Internal local/private enterprise development deployment)